

Machine Learning Notes

1. Introduction to Machine Learning

Machine Learning (ML) is a subfield of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed. Instead of following fixed rules, machine learning algorithms analyze data, identify patterns, and make predictions or decisions.

Key Characteristics of Machine Learning

- Learns automatically from data
- Improves with experience
- Identifies patterns in data
- Makes predictions or decisions

Basic Machine Learning Workflow

1. Data Collection
2. Data Preprocessing
3. Model Training
4. Model Evaluation
5. Prediction or Deployment

Applications of Machine Learning

- Email Spam Detection
- Recommendation Systems
- Fraud Detection
- Medical Diagnosis
- Image Recognition

2. Types of Machine Learning

Machine Learning is mainly divided into three types.

Supervised Learning

Supervised learning uses labeled data where both input and correct output are known. The model learns the relationship between inputs and outputs to predict future data.

Types of Supervised Learning

Regression – Used for predicting numerical values (e.g., house prices).

Classification – Used for predicting categories (e.g., spam or not spam).

Common Algorithms

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors

Unsupervised Learning

Unsupervised learning uses unlabeled data. The algorithm tries to find hidden patterns or groups in the dataset.

Types of Unsupervised Learning

Clustering – Grouping similar data points.

Dimensionality Reduction – Reducing the number of features while keeping important information.

Common Algorithms (Unsupervised)

- K-Means Clustering
- Hierarchical Clustering

- DBSCAN

- PCA

Reinforcement Learning

In reinforcement learning, an agent learns by interacting with an environment and receiving rewards or penalties for actions.

Key Components of Reinforcement Learning

- Agent

- Environment

- Action

- Reward

- Policy

Summary

Supervised Learning – Uses labeled data to predict outputs.

Unsupervised Learning – Finds hidden patterns in unlabeled data.

Reinforcement Learning – Learns optimal actions using rewards.